Roll No							

End Semester Examination 2024

Name of the Course: Diploma CSE

Semester: I V

Name of the Paper: Python Programming

Paper Code: DTCS402

Time: 3 Hour's

Maximum Marks: 100

Note:

(i) All Questions are compulsory.
(ii) Answer any two sub questions among a,b and c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	**(10 X2 = 20 Marks)**	
(a)	State and explain any six features of python. (CO1)	~~CO1,C~~ ~~D2~~
(b)	Explain the concept of variables in Python. Provide examples to illustrate different variable assignments and data types. (CO1)	
(c)	Demonstrate using examples, 4 types of function argument passing methods. (CO2)	
Q2	**(10 X2 = 20 Marks)**	
(a)	Discuss the role of control statements in Python programming. Differentiate between the if, elif, and else statements with examples. (CO2)	CO2
(b)	Discuss List as a fundamental data structure in Python. Explain how to create, access, and modify elements in a list. Provide examples showcasing list operations. (CO2)	
(c)	What is the purpose of the map() function in Python? How do you use the map() function to apply a function to each element of a list? Demonstrate the usage of the map() function with an example? (CO2)	
Q3	**(10 X2 = 20 Marks)**	
(a)	Define what a function is in Python and explain its significance in programming. Provide examples illustrating function definition, arguments, and return values (CO2)	~~CO2,C~~ ~~Q3~~
(b)	Explain the concepts of decorators and generators in Python. Discuss how decorators can modify the behavior of functions and provide examples showcasing their usage. (CO2)	
(c)	Write a Python function to sum all the numbers in a list. (CO3)	
Q4	**(10 X2 = 20 Marks)**	
(a)	Define object-oriented programming (OOP) and its key principles. Explain how classes and objects are used in Python, providing examples of class definition, instantiation, and method invocation. (CO4)	CO4
(b)	Write a Python function using recursion to calculate the factorial of a number (a non-negative integer). The function accepts the number as an argument. (CO4)	
(c)	How do you define and use lambda functions? What are the rules and limitations? Write a lambda function for displaying grater number between two. (CO4)	
Q5	**(10 X2 = 20 Marks)**	
(a)	Describe the purpose of NumPy in Python. Explain how NumPy arrays differ from Python lists. (CO3)	

(b)	Write a program to demonstrate function call using "Passing arguments and returning value". (CO3)	CO3;C 06
(c)	Explain the concept of Object and class in python. What is the difference between class variables and instance variables? (CO5)	